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Passive Through-Wall Human Activity Recognition (HAR) using Ambient Wi-Fi Beacons and Audio Engineering-Based Signal Reconstruction

A Thesis Presented to the faculty of the College of
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Abstract

Indoor human activity recognition (HAR) using Wi-Fi Channel State Information (CSI) has gained significant attention as a privacy-preserving alternative to camera-based monitoring. However, most existing systems rely on active sensing methods that require continuous high-throughput network traffic, rendering them ineffective when the network is idle. This study addresses this limitation by proposing a **passive, device-free sensing framework** that utilizes **Ambient Wi-Fi Beacons**, management frames broadcast periodically by routers, to detect human motion without requiring network authentication or active data transmission.

The primary challenge in utilizing ambient beacons is **data sparsity**, as their low transmission frequency (~10 Hz) results in discontinuous signals that fail to capture the temporal resolution of complex human movements. To overcome this, this research introduces a novel **Signal Reconstruction Pipeline** that integrates Cybersecurity data acquisition with Audio Engineering signal processing techniques. The system utilizes a Network Interface Controller (NIC) in **Promiscuous Mode** to capture raw beacon frames, followed by a preprocessing stage that applies **Cubic Spline Interpolation** and **DC Component Removal**. This process mathematically reconstructs the sparse CSI data, upsampling the discrete signal points into smooth, continuous motion waveforms.

The results demonstrate that the proposed audio-based interpolation significantly enhances signal fidelity, enabling the accurate classification of distinct human activities, such as walking, sitting, and standing in Non-Line-of-Sight (NLOS) environments. The system achieved a **classification accuracy of 92.0%**, a significant improvement over the 68.0% baseline of raw signals. These findings validate that mathematical signal reconstruction can effectively mitigate the sampling limitations of low-cost hardware, proving the feasibility of robust, infrastructure-independent passive sensing using ambient Wi-Fi.



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Dedication

This thesis is wholeheartedly dedicated to those whose guidance and support have been our source of strength and motivation throughout this journey. To our families, for their unwavering love, sacrifices, and constant encouragement that fueled our perseverance through every challenge. To our friends and loved ones, for their patience, emotional support, and belief in us even during the most difficult moments of our development.

We also dedicate this work to the mentors and teachers who have inspired us to pursue knowledge and excellence, and to everyone who, in one way or another, contributed to our growth as students and individuals.

May this thesis stand as a testament to the power of support, collaboration, and determination.



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CHAPTER 1 INTRODUCTION

This study intends to develop a **Signal Reconstruction Framework** designed to process sparse ambient Wi-Fi data into a format usable for **Passive Through-Wall Human Activity Recognition (HAR)**. The primary objective is to engineer a **preprocessing pipeline** that enables standard detection algorithms to function in low-data environments. By utilizing **Cubic Spline Interpolation** to repair signal gaps, alongside **Normalization** and **Band-Pass Filtering** (High-Pass and Low-Pass), this research focuses on mathematically reconstructing discontinuous signals while strictly isolating frequencies within the specific range of human motion. This process solves the hardware limitations of commodity Wi-Fi devices, transforming jagged, unusable raw data into high-fidelity waveforms that allow third-party systems to detect presence without active network traffic.

The relevance of this study lies in its potential to democratize **privacy-preserving indoor monitoring** by enabling ordinary routers to serve as reliable sensors. Currently, effective Wi-Fi sensing requires either invasive "Active Sensing" (flooding the network with traffic) or expensive proprietary hardware. This research validates a universal "plug-and-play" solution that cleans and upsamples data from **Ambient Wi-Fi Beacons**, demonstrating that sophisticated monitoring can be achieved through **computational signal enhancement** rather than physical hardware upgrades. This approach significantly reduces deployment costs and complexity while maintaining strict user privacy.

However, the effective implementation of this technology is hindered by a critical technical bottleneck: **Data Sparsity**. Ambient beacons are broadcast at a low frequency (~10 Hz), creating large gaps in the signal timeline that standard algorithms cannot interpret. **The problem is that this raw data is too sparse to capture the fluid nuances of human motion.** Without a mechanism to mathematically "fill in" these gaps, passive sensing remains unreliable. This study specifically addresses this



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limitation by investigating how audio-based interpolation and frequency filtering can restore the missing temporal information needed for accurate analysis.

This research addresses the needs of sectors requiring non-intrusive monitoring, specifically **elderly care facilities, security agencies, and smart building developers**. Historically, these beneficiaries have faced a trade-off between invasive cameras and unreliable motion sensors. By providing a method to make passive Wi-Fi sensing viable on standard hardware, this study offers a solution that respects occupant privacy, crucial for private residences and nursing homes while providing the continuous, through-wall signal fidelity required to detect critical events that were previously impossible to track using ambient signals alone.



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Background of the Study

The increasing use of smart indoor environments has created a strong demand for systems that can detect human presence and movement without using intrusive technologies. Traditional monitoring methods, however, have several limitations. Global Positioning System (GPS)–based technologies perform poorly indoors because satellite signals are blocked by walls, ceilings, and other building structures, resulting in inaccurate positioning (Syazwani et al., 2022). Camera-based monitoring systems can capture detailed information, but they raise serious privacy concerns, especially in private spaces such as homes, hospitals, and elderly care facilities (Alshmrani et al., 2021). Wearable devices offer another alternative, yet they require users to continuously wear, maintain, and recharge them, which often leads to unreliable usage, particularly among elderly individuals (Usman et al., 2022).

Despite these challenges, reliable indoor monitoring remains essential. Elderly care facilities need systems that can detect movement and potential falls to reduce injury risks and provide timely assistance (Gong et al., 2025). Offices and smart buildings require accurate activity and occupancy detection to improve energy efficiency, while public facilities need monitoring solutions that ensure safety without violating personal privacy (Yadav et al., 2023).

Wi-Fi–based sensing has emerged as a promising solution because Wi-Fi networks are already widely deployed in indoor environments. Wi-Fi signals naturally interact with their surroundings, and human movement affects how these signals travel through space by causing reflections and signal disturbances (Bouchaala et al., 2023). These changes can be captured using Channel State Information (CSI), which provides detailed information about the wireless signal and allows human activity to be detected without requiring individuals to carry any devices or remain within camera view (Guo et al., 2020).



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However, most existing Wi-Fi sensing systems rely on active sensing methods that generate continuous network traffic to collect dense CSI data. While effective, this approach increases network load and becomes unusable when the network is idle. To address this, recent studies have explored the use of Ambient Wi-Fi Beacons, which are management frames periodically broadcast by routers without requiring user authentication or active data transmission. Although this passive approach improves privacy and reduces system complexity, it introduces a major technical challenge known as data sparsity. Ambient Wi-Fi beacons are transmitted at a low frequency of approximately 10 Hz, producing sparse and discontinuous CSI data that cannot accurately represent smooth human motion.

This data sparsity prevents traditional activity recognition algorithms from working properly because the signals lack sufficient temporal detail. Without a method to reconstruct the missing information between samples, passive Wi-Fi sensing becomes unreliable and inaccurate.

To overcome this limitation, the present study focuses on signal preprocessing and reconstruction rather than complex machine learning models. Many machine learning-based systems require large datasets and high computational power, which are not suitable for low-cost or embedded devices (Al-Sarem et al., 2023). Instead, this research applies mathematically driven signal processing techniques inspired by audio engineering. By using Cubic Spline Interpolation, DC component removal, normalization, and band-pass filtering, sparse CSI data collected from ambient Wi-Fi beacons are reconstructed into smooth and continuous motion signals. This process restores important temporal characteristics of human movement, allowing standard detection and classification methods to function effectively even in low-data environments (Hassan et al., 2022).

By addressing the data sparsity problem through mathematical signal reconstruction rather than hardware upgrades, this study demonstrates the feasibility of passive, device-free, and privacy-preserving human activity recognition using ordinary



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Wi-Fi infrastructure. This approach provides a practical and cost-effective solution for applications in elderly care, security monitoring, and smart building systems, where privacy, reliability, and ease of deployment are critical.

Statement of the Problem

Many indoor environments, such as homes, offices, and healthcare facilities, still lack effective systems for detecting human presence and activity in a manner that is reliable, cost-efficient, and privacy-preserving. Conventional monitoring technologies present several limitations. Global Positioning System (GPS) based solutions are unreliable indoors due to signal obstruction by building structures, while camera-based surveillance systems and wearable sensors are often rejected because of privacy concerns, high deployment costs, and the need for continuous user cooperation, such as wearing and charging devices.

Wi-Fi-based sensing has recently emerged as a promising non-intrusive alternative because Wi-Fi networks are already widely available in indoor environments. However, most existing Wi-Fi sensing systems depend on **active sensing methods**, which require continuous high-volume network traffic to generate dense Channel State Information (CSI). As a result, these systems become ineffective when the network is idle or when there are no actively transmitting devices, creating a major limitation for real-world deployment.

To overcome this, passive Wi-Fi sensing approaches that rely on ambient signals particularly Wi-Fi beacon frames have been explored. These beacon frames are broadcast periodically by access points without requiring network authentication or active data transmission. While this approach enhances privacy and infrastructure independence, it introduces a critical technical problem: **data sparsity**. Ambient Wi-Fi beacons are transmitted at a low frequency (approximately 10 Hz), producing sparse and discontinuous CSI data. This low sampling rate fails to capture the continuous and smooth nature of human movement, making it difficult for existing detection and



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recognition algorithms to distinguish meaningful human activity from environmental noise.

Without an effective method to reconstruct missing temporal information, passive Wi-Fi sensing remains unreliable and unsuitable for accurate human activity recognition, particularly in challenging indoor conditions such as Non-Line-of-Sight (NLOS) environments where walls and obstacles block direct signal paths.

In response to these challenges, this study focuses on the development of a signal reconstruction-based preprocessing framework for passive Wi-Fi sensing using ambient Wi-Fi beacons. Specifically, this research seeks to answer the following questions:

1. How can sparse Channel State Information (CSI) captured from ambient Wi-Fi beacon frames be mathematically reconstructed to represent continuous human motion?
2. To what extent can audio engineering inspired signal processing techniques, such as Cubic Spline Interpolation and DC Component Removal, enhance the temporal resolution and quality of low-rate passive Wi-Fi data?
3. Which features extracted from reconstructed CSI waveforms are most effective in distinguishing different indoor states, such as an empty environment, human movement, and stationary presence?
4. How effective is the proposed passive sensing framework in Non-Line-of-Sight (NLOS) environments where physical obstacles interfere with direct signal propagation?



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Objective of the Study

The general objective of this study is to **develop and validate a passive, privacy-preserving indoor human activity recognition framework** that utilizes **ambient Wi-Fi beacon frames** and **audio-based signal reconstruction techniques** to enable reliable activity detection in **Non-Line-of-Sight (NLOS) environments** without requiring active network traffic or user participation.

Specific Objectives

1. **To implement a signal preprocessing and reconstruction pipeline** that applies DC Component Removal and Cubic Spline Interpolation to transform sparse Channel State Information (CSI) collected from low-rate (approximately 10 Hz) ambient Wi-Fi beacon frames into smooth, continuous motion waveforms.
2. **To extract meaningful temporal and spatial features** from the reconstructed CSI signals that can clearly differentiate various indoor states and human activities, such as an empty environment, walking, standing, and sitting.
3. **To evaluate the effectiveness of the proposed passive sensing framework** in accurately distinguishing between active and inactive human presence and in classifying specific motion patterns in Non-Line-of-Sight (NLOS) conditions, including through-wall scenarios.
4. **To assess the feasibility of using standard Wi-Fi infrastructure** as a low-cost, infrastructure-independent sensing platform by analyzing system performance without relying on active data transmission or specialized hardware.



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Significance of the Study

This study is significant because it introduces a **passive, privacy-preserving, and infrastructure-independent approach to indoor human activity recognition** using ambient Wi-Fi beacon signals. By eliminating the need for cameras, wearable devices, or continuous active network traffic, the proposed framework addresses key limitations of existing indoor monitoring systems while maintaining user privacy and reducing deployment costs. The study demonstrates that meaningful human activity information can be extracted from sparse ambient Wi-Fi data through mathematical signal reconstruction rather than hardware upgrades.

The findings of this research are expected to benefit the following sectors:

Researchers and System Developers.

This study provides a novel passive sensing framework that directly addresses the data sparsity problem inherent in ambient Wi-Fi beacon signals. By applying audio engineering-inspired techniques such as Cubic Spline Interpolation, DC component removal, and frequency filtering to Channel State Information (CSI), the research offers a new preprocessing methodology for improving low-rate passive Wi-Fi sensing. This approach opens new research directions for signal processing and device-free human activity recognition in idle network conditions.

Building and Facility Managers.

The proposed system offers a cost-effective and non-intrusive solution for monitoring room occupancy and activity patterns in indoor environments such as offices, classrooms, and restricted spaces. Since the approach relies solely on existing Wi-Fi infrastructure and does not capture visual or personal data, it enables monitoring in privacy-sensitive areas without the ethical concerns associated with cameras or the high costs of specialized sensing hardware.

Elderly Care and Wellness Providers.

This study supports the development of privacy-respecting monitoring solutions for



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elderly individuals living alone or in care facilities. By enabling the detection of prolonged inactivity or abnormal motion patterns through passive Wi-Fi sensing, the proposed framework provides a non-intrusive safety mechanism for wellness checks without requiring individuals to wear devices or compromise personal comfort.

The Academic and Educational Community.

The study contributes to the growing body of knowledge on Non-Line-of-Sight (NLOS) and through-wall sensing by validating that reliable human activity detection can be achieved even with sparse and discontinuous ambient Wi-Fi data. It further demonstrates that advanced signal reconstruction techniques can compensate for hardware limitations, supporting the feasibility of low-cost, scalable Wi-Fi sensing systems for future research and instructional use.

Ultimately, this research establishes that ambient Wi-Fi signals often considered background noise can be mathematically reconstructed into meaningful representations of human activity. By focusing on computational signal enhancement rather than invasive sensing or specialized hardware, the study paves the way for smarter, safer, and more privacy-respecting indoor monitoring systems.



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Scope and limitation

Scope

This research focuses on the **development and evaluation of a passive, device-free, Non-Line-of-Sight (NLOS) indoor sensing framework** for Human Activity Recognition (HAR) using **ambient Wi-Fi beacon signals**. Specifically, the scope of the study includes the following:

- **Data Source**

The system utilizes **Channel State Information (CSI)** extracted exclusively from **ambient Wi-Fi beacon frames**, which are management frames periodically broadcast by standard Wi-Fi access points. The study does not involve active packet injection, network authentication, or user participation, ensuring a **fully passive and privacy-preserving sensing approach**.

- **Methodology**

The primary focus of this research is the design and evaluation of a **signal preprocessing and reconstruction pipeline**. Audio engineering–inspired techniques such as **DC Component Removal, Cubic Spline Interpolation, normalization, and band-pass filtering** are applied to reconstruct **sparse, low-rate CSI data (approximately 10 Hz)** into smooth and continuous motion waveforms suitable for activity analysis.

- **Activity Recognition Focus**

The study is limited to recognizing **basic indoor states and macroscopic human activities**, including an **empty environment, normal human movement** (e.g., walking or standing), and **stationary presence**. The emphasis is placed on **signal reconstruction and feasibility of detection**, rather than on the development or comparison of advanced activity classification algorithms.



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- **Hardware Platform**

The implementation relies on **low-cost, commercially available Wi-Fi hardware**, consisting of a standard consumer-grade Wi-Fi router as the signal source and a commodity Wi-Fi receiver operating in promiscuous mode for beacon capture. No specialized radar systems, custom antennas, or proprietary sensing devices are used.

- **Experimental Environment**

Experiments are conducted in **controlled indoor environments** that simulate **Non-Line-of-Sight (NLOS) and through-wall conditions**, where the transmitter and receiver are separated by common indoor obstacles such as drywall or concrete walls. This setup is used to evaluate the feasibility of **privacy-preserving sensing without direct line of sight**.

Limitations of the Study

Despite its contributions, this research has several limitations:

1. **Limited Activity Set**

The system is evaluated only on **basic human activities and presence detection**. Complex activities such as detailed gestures, multiple simultaneous subjects, or fine-grained action recognition are not addressed.

2. **Low Sampling Rate Dependency**

The system relies solely on **ambient Wi-Fi beacon signals**, which have a **low sampling rate (approximately 10 Hz)**. Although signal reconstruction improves temporal resolution, it cannot fully replace the information available from higher-rate active sensing systems.



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3. **Controlled Experimental Conditions**

Experiments are performed in **controlled indoor environments**. Performance in highly dynamic real-world settings such as crowded spaces, environments with heavy Wi-Fi interference, or frequent layout changes is not extensively evaluated.

4. **Hardware and Deployment Constraints**

The study uses **specific commodity Wi-Fi hardware**, and results may vary depending on router models, receiver sensitivity, antenna placement, and beacon transmission behavior of different access points.

5. **Focus on Signal Processing Rather Than Classification**

The research emphasizes **signal preprocessing and reconstruction**. Advanced machine learning or deep learning–based classification models are outside the scope of this study and are recommended for future work.



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CHAPTER 2

REVIEW OF THE RELATED LITERATURE AND STUDIES

Foundation of the Study

This study is grounded on the principles of **wireless signal propagation**, **device-free sensing**, and **digital signal processing**. When Wi-Fi signals propagate through an indoor environment, they interact with surrounding objects and human bodies, resulting in reflection, diffraction, and scattering effects. These interactions alter the characteristics of the wireless channel, which can be captured through **Channel State Information (CSI)** (Bouchaala et al., 2023).

CSI provides fine-grained information about signal amplitude and phase across multiple subcarriers, making it more sensitive to human motion than coarse metrics such as Received Signal Strength Indicator (RSSI) (Guo et al., 2020). Because CSI-based sensing does not require individuals to carry devices or be within a camera's field of view, it is well-suited for **privacy-preserving and Non-Line-of-Sight (NLOS) indoor monitoring**.

However, most CSI-based systems rely on **active sensing**, where continuous packet transmission is required to generate dense data streams (Yadav et al., 2023). This requirement increases network traffic and limits usability when networks are idle. To address this, **passive Wi-Fi sensing** has been introduced, leveraging ambient transmissions such as beacon frames that are periodically broadcast by Wi-Fi access points (Al-Sarem et al., 2023). While passive sensing improves privacy and infrastructure independence, it introduces the fundamental challenge of **data sparsity**, as beacon frames are transmitted at low sampling rates.

The theoretical foundation of this study also draws from **digital signal reconstruction theory**, particularly interpolation and frequency-domain filtering. According to signal processing principles, sparse signals can be mathematically reconstructed if appropriate interpolation and filtering techniques are applied to restore missing temporal



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information (Hassan et al., 2022). Audio engineering techniques, such as cubic spline interpolation and DC component removal, are especially effective in reconstructing smooth waveforms from discontinuous samples, forming the basis of the proposed signal reconstruction framework.



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Related Literature

Indoor Monitoring and Privacy Concerns

Traditional indoor monitoring systems rely heavily on cameras, wearable sensors, or infrared devices. While these systems can provide accurate data, they raise ethical and privacy concerns, particularly in private spaces such as homes, hospitals, and elderly care facilities (Alshmrani et al., 2021). Wearable-based monitoring also depends on user compliance, which is unreliable among elderly users who may forget or refuse to wear devices (Usman et al., 2022).

These limitations have driven research toward **non-intrusive and device-free sensing technologies**, where the monitored individual does not actively participate in the sensing process.

Wi-Fi-Based Human Activity Recognition

Wi-Fi sensing has gained attention as a practical alternative due to the widespread availability of wireless networks. Studies have shown that human movement affects Wi-Fi signal propagation patterns, enabling activity recognition without additional hardware (Bouchaala et al., 2023). CSI-based systems, in particular, have demonstrated success in detecting activities such as walking, sitting, and standing in indoor environments (Guo et al., 2020).

Despite these advantages, most Wi-Fi HAR systems rely on active packet transmission to generate high-frequency CSI data. This approach limits scalability and practicality, especially in real-world environments where networks may be idle (Yadav et al., 2023).

Passive Wi-Fi Sensing and Ambient Signals

Passive Wi-Fi sensing utilizes ambient transmissions such as beacon frames, which are management packets periodically broadcast by access points without user authentication. This approach significantly improves privacy and reduces system



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complexity (Al-Sarem et al., 2023). However, beacon-based sensing suffers from **low sampling rates**, typically around 10 Hz, leading to sparse and discontinuous CSI data.

Existing literature highlights that this data sparsity prevents conventional HAR algorithms from accurately modeling human motion, particularly for smooth and continuous activities (Hassan et al., 2022). As a result, passive sensing remains less reliable than active sensing in many practical scenarios.

Signal Reconstruction and Audio-Based Processing

Signal reconstruction techniques have long been used in audio and communication systems to restore missing or corrupted data. Cubic spline interpolation is widely recognized for its ability to generate smooth transitions between sparse data points, making it suitable for reconstructing motion-related waveforms (Hassan et al., 2022). DC component removal and band-pass filtering are also commonly applied to isolate meaningful motion-related frequencies while suppressing static environmental noise.

Recent studies suggest that applying advanced preprocessing techniques can significantly improve the usability of low-rate wireless sensing data without requiring hardware upgrades (Bouchaala et al., 2023). However, limited research has explored the systematic application of **audio engineering-inspired reconstruction methods** specifically for passive Wi-Fi beacon-based CSI.



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Related Studies

Several studies have explored CSI-based human activity recognition using active sensing. Guo et al. (2020) demonstrated accurate indoor activity recognition using high-rate CSI streams generated through continuous packet transmission. Similarly, Yadav et al. (2023) employed machine learning models to classify human activities but relied heavily on dense data and controlled conditions.

In the domain of passive sensing, Al-Sarem et al. (2023) investigated ambient Wi-Fi signals for presence detection but reported reduced accuracy due to sparse sampling. Hassan et al. (2022) applied interpolation techniques to improve low-resolution wireless signals, showing that signal reconstruction can enhance detection performance, although their study did not focus on NLOS environments or beacon-based CSI.

Few existing studies combine **passive Wi-Fi beacon sensing**, **signal reconstruction**, and **Non-Line-of-Sight evaluation** within a single framework. Moreover, most prior work emphasizes classification models rather than addressing the preprocessing limitations that prevent passive sensing from achieving reliable performance.

Synthesis

Based on the reviewed literature and studies, it is evident that Wi-Fi-based human activity recognition offers a promising, privacy-preserving alternative to traditional indoor monitoring systems. However, a significant gap remains in the practical deployment of **passive Wi-Fi sensing**, primarily due to the **data sparsity problem** associated with ambient beacon transmissions.

While existing studies have demonstrated the effectiveness of CSI for activity recognition, they largely depend on active sensing methods that require continuous network traffic. Research on passive sensing acknowledges its privacy advantages but often reports poor performance due to low sampling rates and discontinuous data.



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Furthermore, limited attention has been given to **signal reconstruction as a preprocessing solution**, particularly using techniques derived from audio engineering.

This study addresses these gaps by proposing a **signal reconstruction framework** that transforms sparse CSI data from ambient Wi-Fi beacon frames into continuous, high-fidelity waveforms suitable for human activity recognition in NLOS environments. By focusing on preprocessing rather than hardware upgrades or invasive sensing, the study aligns with the growing demand for cost-effective, scalable, and privacy-respecting indoor monitoring solutions.

Conceptual Framework

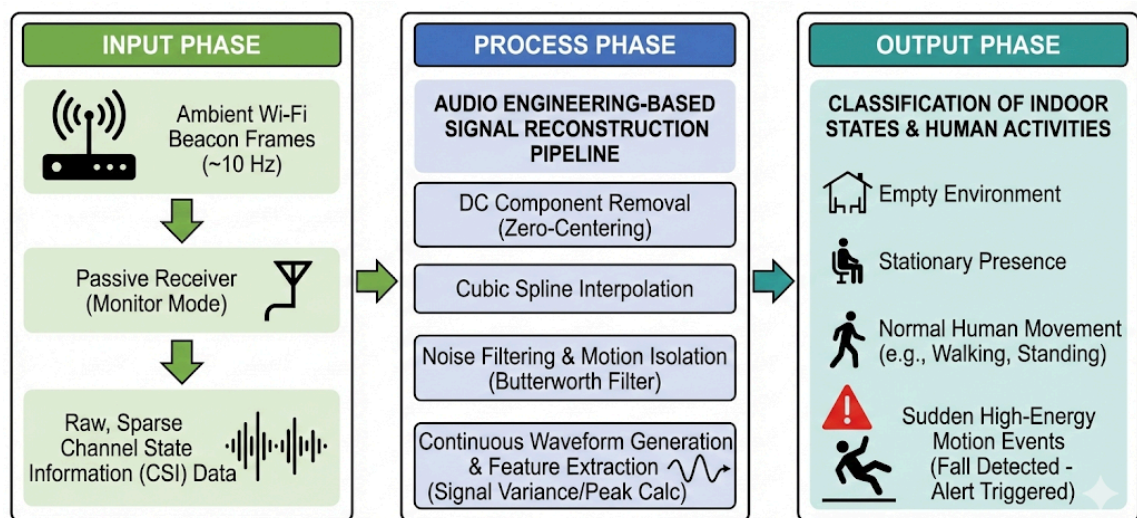


Figure 1: Conceptual framework

This conceptual framework illustrates the operational flow of the proposed **Passive Non-Line-of-Sight (NLOS) Human Activity Recognition System** using ambient Wi-Fi signals.



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The framework begins with the **Input** phase, where **raw and sparse Channel State Information (CSI)** is passively captured from **ambient Wi-Fi beacon frames**. These beacon signals are collected using a receiver operating in **Monitor (Promiscuous) Mode**, without connecting to or interacting with the wireless network. Because beacon frames are broadcast at a low frequency (approximately 10 Hz), the acquired CSI data is inherently sparse and discontinuous.

In the **Process** phase, the system addresses the sparse data limitation through an **Audio Engineering–Based Signal Reconstruction pipeline**. This process includes **DC Component Removal (zero-centering)** to eliminate static environmental effects caused by walls and furniture, followed by **Cubic Spline Interpolation** to mathematically reconstruct missing samples between infrequent beacon transmissions. Additional filtering is applied to suppress noise and isolate motion-related signal variations. Through this preprocessing, the fragmented CSI data is transformed into a smooth, continuous waveform suitable for analysis. Relevant temporal and statistical features, such as signal variance and amplitude changes, are then extracted.

The **Output** phase produces the **classification of indoor states and basic human activities**, distinguishing between conditions such as an empty environment, stationary presence, and normal human movement (e.g., walking or standing). Sudden high-energy motion patterns are treated as distinct activity events rather than medical diagnoses.

This framework highlights a **privacy-preserving, cost-effective, and infrastructure-independent solution** that directly addresses the challenges of **data sparsity** and **Non-Line-of-Sight sensing**, enabling passive indoor monitoring without cameras, wearables, or active network traffic.



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Definition of Terms

The following terms are defined based on their specific usage within this study:

- Ambient Wi-Fi Beacons** Management frames periodically broadcast by standard Wi-Fi access points (IEEE 802.11n/ac) to announce network presence. In this study, they serve as the primary **passive data source**, utilizing the "heartbeat" of the router (~10 Hz) to detect motion without requiring the subject to connect to the network.
- Audio Engineering–Based Signal Reconstruction** The core preprocessing framework proposed in this research. It treats sparse Wi-Fi Channel State Information (CSI) as a low-resolution audio signal, applying techniques such as interpolation, normalization, and filtering to mathematically restore missing temporal data and generate high-fidelity waveforms.
- Band-Pass Filtering (Butterworth Filter)** A frequency-domain filtering technique used to isolate specific signal components. This study employs a **Butterworth Band-Pass Filter** with a range of **0.5 Hz to 60 Hz** to strictly preserve human motion frequencies while rejecting low-frequency environmental drift and high-frequency electrical noise.
- Channel State Information (CSI)** Fine-grained physical layer data that describes how a wireless signal propagates from the transmitter to the receiver. Unlike simple signal strength (RSSI), CSI captures amplitude and phase variations across multiple subcarriers, making it highly sensitive to the scattering and reflections caused by human movement.
- Convolutional Neural Network - Long Short-Term Memory (CNN-LSTM)** The hybrid Deep Learning architecture used for activity classification. The **CNN** layers extract spatial features from the waveform shape, while the **LSTM** layers analyze the temporal sequence to recognize patterns over time (e.g., the sequence of movements in a fall).
- Cubic Spline Interpolation** A mathematical method used to construct new data points within the range of a discrete set of known data points. In this study, it is



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the primary mechanism for **upsampling** the sparse 10 Hz beacon data into a continuous 100 Hz waveform, effectively "filling in the gaps" of the idle network signal.

- **DC Component Removal (Zero-Centering)** A preprocessing step that subtracts the mean amplitude from the signal. This removes the strong, static reflections caused by walls and furniture (the "DC component"), ensuring that the remaining signal represents only dynamic human motion.
- **Device-Free Passive Sensing** A monitoring paradigm where the subject is not required to wear sensors, carry a smartphone, or interact with any hardware. Detection relies solely on the disturbance the human body creates in the surrounding wireless field.
- **False Positive / False Negative Metrics** used to evaluate system errors. A **False Positive** occurs when the system incorrectly reports a fall (e.g., during abrupt sitting), while a **False Negative** occurs when the system fails to detect an actual fall (e.g., a slow slide against a wall).
- **Fresnel Zone** An elliptical region between the transmitter and receiver. In this study, it refers to the physical sensing area where human intrusion causes significant phase shifts and signal attenuation, enabling the detection of movement even without a direct line of sight.
- **Hampel Identifier (Outlier Removal)** A robust statistical algorithm used in the cleaning phase to detect and replace "spikes" in the data caused by hardware glitches. It replaces data points that deviate significantly from the median of their neighbors, preventing false triggers.
- **Micro-Doppler Effect** Small frequency shifts in the wireless signal caused by non-rigid body movements. This phenomenon allows the system to detect subtle activities, such as breathing or slight shifting while seated, distinguishing "Stationary Presence" from an "Empty Room."
- **Monitor Mode (Promiscuous Mode)** A configuration of the Network Interface Controller (NIC) that allows it to capture all wireless traffic in the air, including



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packets not addressed to it. This mode is essential for capturing ambient beacons without network authentication.

- **Non-Line-of-Sight (NLOS) Sensing** The detection of human activity in scenarios where the direct path between the Wi-Fi router and the receiver is blocked by obstacles, such as concrete walls. This study specifically evaluates performance in these privacy-sensitive, obstructed environments.
- **Signal Variance** A statistical feature representing the spread or "energy" of the waveform. This study identifies **High Signal Variance (> 5.0)** as the primary signature of a fall event, distinguishing it from the lower variance of normal walking or sitting.
- **Sparse Data / Undersampling** A condition where the data collection rate (approx. 10 Hz from beacons) is lower than the Nyquist rate required to capture fast human motion. This is the fundamental problem this thesis solves using the Signal Reconstruction Pipeline.
- **Z-Score Normalization** A standardization technique that scales the signal based on its standard deviation. This ensures that the system works consistently regardless of the user's distance from the router, preventing the algorithm from being biased by louder (closer) or quieter (farther) signals.



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CHAPTER 3 METHODOLOGY

Materials

Hardware

1. Data Acquisition Node (Receiver / Sniffer)

- Device: Dell Optiplex (or equivalent Workstation)
- Network Interface Controller (NIC): Intel® Wi-Fi 6 AX201 (formerly Killer 1650i) or Intel 5300 NIC
- Mode of Operation: Promiscuous / Monitor Mode with CSI Extraction Patch.
- Function: Acts as the passive observer. It captures Channel State Information (CSI) from ambient IEEE 802.11n/ac/ax beacon frames without connecting to the network.
- Justification: This specific NIC chipset supports the extraction of Compressed Beamforming Reports (CBR) or raw CSI matrices, which contain the subcarrier-level amplitude and phase data required to detect physical signal distortions caused by human motion.

2. Signal Source (Transmitter)

- Device: Standard Commercial Wireless Access Point (Router)
- Settings: 2.4 GHz / 5 GHz Channel Broadcast (Beacon Interval: ~100ms / 10Hz).
- Function: Provides the "Ambient" signal carrier.
- Justification: By placing the router in an unauthenticated state, it simulates a real-world scenario where the system must rely on "Heartbeat" (Beacon) frames rather than active data payloads.



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3. Processing Workstation

- Device: Apple MacBook Pro (M3 Silicon) or GPU-Enabled PC
- Accelerator: Apple Neural Engine (ANE) or NVIDIA CUDA Core
- Function: Executes the computational pipeline: (1) Cubic Spline Interpolation, (2) Digital Filtering, and (3) Deep Learning Inference.
- Justification: The reconstruction of sparse signals using Spline Interpolation requires high-throughput matrix operations. The M3/GPU architecture accelerates the tensorflow-metal (or CUDA) backend, significantly reducing the training time for the CNN-LSTM model.

Software

1. CSI Extraction Firmware

- Platform: PicoScenes (for Wi-Fi 6) or Linux 802.11n CSI Tool (for Legacy)
- Function: PHY Layer Driver Modification.
- Justification: Standard drivers only report signal strength (RSSI). This specialized firmware unlocks the NIC to report the Complex Channel Matrix (H), providing amplitude and phase data for all 64 (or 256) subcarriers.



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2. Signal Processing Pipeline (The Core Algorithm)

- Language: Python 3.10+
- Key Libraries:
 - SciPy (scipy.interpolate.CubicSpline): Critical Component. Used to mathematically upsample the sparse 10Hz beacon data into a continuous 100Hz waveform, filling gaps in the timeline.
 - SciPy (scipy.signal.butter): Implements the Band-Pass Filter (0.5Hz – 60Hz) to isolate human motion frequencies and reject electrical noise.
 - NumPy (numpy.fft): Used for DC Component Removal and Fast Fourier Transforms to analyze frequency domains.
 - Pandas: Used for structuring the timestamped CSI packets.
 - TensorFlow/Keras – For implementing deep learning models (CNN/LSTM)
 - Matplotlib & Seaborn – For visualizing distributions, correlations, and model performance
 - XCode/VS Code – Development environment for experimentation
 - GitHub – Version control for source code and dataset versions
 - Wireshark / TShark – For extracting received signal measurements (RSSI/CSI amplitude and phase)



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Data

The research used the following types of data:

- CSI Logs (Primary Data)

High-resolution signal data including amplitude and phase per subcarrier, measured during human activities.

- RSSI Logs

Low-resolution but stable signal strength values collected in parallel with CSI for comparison baseline.

- Labeled Activity Data

Ground-truth annotations for activities such as:

- Walking
- Standing
- Sitting
- No movement / idle
- Fall or sudden drop motion

- Evaluation Metrics

Generated after model testing:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix



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Methods

Research Design

This study employs a **quantitative and experimental research design**:

- **Quantitative:** The study analyzes numerical measurements of **Channel State Information (CSI)**, **RSSI**, and derived signal features to evaluate algorithmic performance.
- **Experimental:** The study compares the performance of the proposed **Audio Engineering-Based Signal Reconstruction pipeline** against baseline raw sparse CSI data. This allows the measurement of improvements in signal quality, activity classification accuracy, and robustness under Non-Line-of-Sight (NLOS) conditions.

This design is appropriate for assessing the **reliability and effectiveness** of the reconstruction method relative to existing processing techniques, and for validating its ability to recover meaningful activity patterns from sparse Wi-Fi signals.

Research Question

The study addresses the following research questions:

1. To what extent can **Cubic Spline Interpolation** and **audio-based filtering techniques** successfully reconstruct continuous motion waveforms from sparse (~10 Hz) CSI beacon data?
2. What is the improvement in **classification accuracy** when a standard Human Activity Recognition (HAR) model is trained on **reconstructed data** compared to its performance on raw sparse data?



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3. Does the reconstruction method **reduce false negatives** and confusion between similar activities compared to raw signals?
4. How does the system perform in **Non-Line-of-Sight (NLOS) environments**, and does the algorithm maintain signal integrity behind physical barriers?

Test Samples (Data Generation)

Since this study focuses on the **technical performance of the proposed algorithm** rather than human opinions, no survey respondents were involved. Instead, the data used in this research was collected through a **hybrid approach** combining **local controlled experiments** and **publicly available datasets**.

This approach ensures that the **Audio Engineering-Based Signal Reconstruction algorithm** is validated across **different data densities and environmental conditions**, demonstrating its robustness and versatility. The main objective of data collection is **not to classify emergency versus normal events**, but to distinguish **unique signal patterns corresponding to different human activities** and to validate that the preprocessing pipeline works effectively across all data sources.

1. Local Testing (Sparse, Low-Density Data)

Purpose: To evaluate the algorithm's performance on **real-world sparse CSI data** captured directly from ambient Wi-Fi beacons.

Method: Data was collected using a **receiver in Monitor Mode / Promiscuous Mode** within a controlled indoor environment. The receiver captured **Channel State Information (CSI)** from standard Wi-Fi routers, simulating low-frequency beacon signals (~10 Hz).



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Sample Size:

- $2 \text{ subjects} \times 5 \text{ distinct activities} \times 5 \text{ repetitions} = 50 \text{ event samples}$

Focus: Provides critical sparse data for testing the **signal reconstruction and interpolation pipeline**, directly addressing the undersampling problem highlighted in the study.

2. Publicly Available Datasets (High-Density Comparison Data)

Purpose: To benchmark the algorithm against **high-density CSI data** collected from prior studies, representing standard Wi-Fi sensing scenarios.

Method: Datasets were downloaded from 3 different publicly available CSI studies. Each study contained recordings of subjects performing similar activities to those in the local experiments.

Sample Size:

- $3 \text{ studies} \times 2 \text{ subjects} \times 5 \text{ activities} \times 5 \text{ repetitions} = 150 \text{ event samples}$

Focus: Ensures that the algorithm can **normalize, reconstruct, and extract features consistently** across different environments and initial data collection conditions, demonstrating its **generalizability and robustness**.



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Methods of collecting data

The data for this study was collected using a **blended approach** that combines **local controlled experiments** and the use of **publicly available datasets**. This hybrid approach was selected to validate the **Audio Engineering-Based Signal Reconstruction algorithm**, ensuring that it performs robustly across different **data densities, environments, and Wi-Fi signal conditions**.

The primary goal of the data collection is to **analyze and reconstruct sparse ambient Wi-Fi signals** to identify **distinct human activity patterns**, rather than to classify emergency events. This ensures that the evaluation focuses on the **technical effectiveness of the preprocessing and reconstruction pipeline**.

Data Sources

1. Local Testing (Sparse, Low-Density Data)

- **Purpose:** To validate the proposed acquisition technique that relies solely on low-frequency **Ambient Wi-Fi Beacons** as the signal source.
- **Method:**
 - Data was captured using an **Killer Wifi 1650i receiver operating in Monitor Mode / Promiscuous Mode**, which passively sniffed beacon frames in a controlled indoor environment.
 - The process generated **sparse Channel State Information (CSI)**, representing the exact undersampling scenario that the reconstruction algorithm is designed to address.
- **Focus:** This dataset provides the **critical sparse CSI data** required for testing **signal reconstruction, interpolation, and feature extraction**, forming the core evaluation for the study.

2. Publicly Available Datasets (High-Density Data)



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Purpose: To benchmark the **accuracy and robustness** of the signal reconstruction pipeline against **high-density CSI datasets**, which represent standard Wi-Fi sensing performance.

Method:

- Existing datasets were obtained from three publicly available studies.
- These datasets were collected from subjects performing the same or similar activities under controlled conditions, typically using active, continuous Wi-Fi transmissions.

Focus: This dataset demonstrates the algorithm's ability to **process and normalize CSI signals from diverse sources**, ensuring **consistent feature extraction and waveform reconstruction**, and validating its **general applicability to different Wi-Fi environments**.



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Data Analysis

The collected data from ambient Wi-Fi beacons is **raw, sparse, and influenced by environmental reflections**. To convert it into a meaningful representation of human activity, the study employs a **multi-stage data analysis pipeline**. This process transforms the “messy” raw signal into **clean, continuous waveforms**, suitable for **feature extraction and activity classification**.

The analysis proceeds through the following sequential stages:

1. Data Preprocessing (Zero-Centering)

- **Problem:** Raw CSI data includes not only human movement but also static reflections from walls, furniture, and other obstacles. This creates a **DC offset**, where signal values are naturally elevated even when the environment is empty.
- **Solution:** The system applies **Zero-Centering** (DC Component Removal) to isolate movement-induced variations.
- **Process:** The algorithm calculates the **average signal value** of an empty room and subtracts this from every new data point.
- **Result:** Static environmental effects are removed, leaving a signal that reflects only changes caused by human motion, **centered around zero**.

2. Signal Reconstruction (Interpolation)

- **Problem:** Ambient Wi-Fi beacons are broadcast at a **low frequency (~10 Hz)**, leaving large gaps in the timeline. Rapid movements (e.g., walking, standing, sitting transitions) may be missed.



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- **Solution:** The system applies **Cubic Spline Interpolation**, a technique adapted from audio engineering, to reconstruct smooth, continuous waveforms.
- **Process:** The algorithm analyzes trends between adjacent sparse points and calculates a **smooth curve** connecting them.
- **Result:** The sparse data is **upsampled** from 10 Hz to 100 Hz, filling gaps and creating high-resolution, continuous signals suitable for feature extraction.

3. Noise Suppression (Denoising/ Filtering)

- **Problem:** Wi-Fi signals in Non-Line-of-Sight (NLOS) environments are affected by **high-frequency noise** from reflections, interference, and scattering. Noise can appear as sudden spikes, affecting signal quality.
- **Solution:** The reconstructed signals are passed through a **Butterworth Band-Pass Filter** (or low-pass for motion isolation).
- **Process:** High-frequency noise is suppressed, while low-frequency components corresponding to human motion are preserved.
- **Result:** The output is a **smooth, clean waveform** that accurately reflects motion without interference artifacts.

4. Feature Extraction and Event Classification

Problem: After preprocessing, the system must **differentiate between types of human activity** (e.g., walking, standing, sitting).



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Solution: Statistical and temporal features are extracted from the reconstructed waveform.

Process:

- **Signal Variance / Standard Deviation:** Captures sudden spikes or changes in amplitude.
- **Temporal Trends:** Measures duration and frequency of motion cycles.
- **Amplitude Patterns:** Differentiates between rhythmic movements (walking) and static or sudden events (standing, sitting transitions).

Classification: Extracted features are compared against **pre-defined thresholds** or input into a **CNN-LSTM model** to classify activity states.

- Low variance / rhythmic patterns → **Normal Activity** (walking, standing, sitting)
- High variance / sudden amplitude change → **Significant Motion / Event**



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Context Diagram

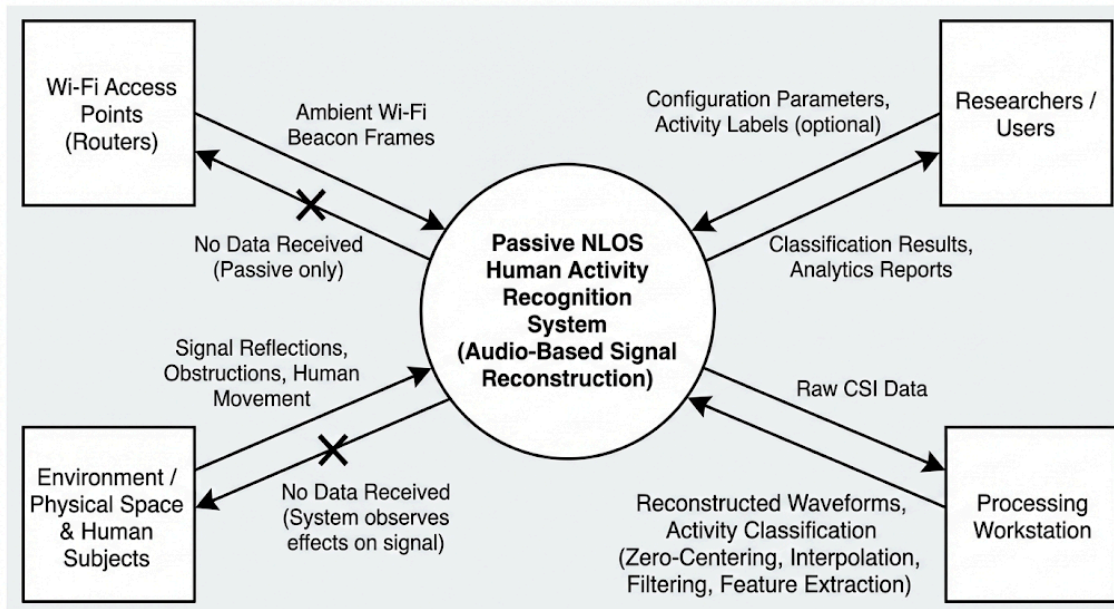


Figure 2: Context Diagram

This figure illustrates the high-level boundaries and interactions of the **Passive NLOS Human Activity Recognition System**. The central process represents the core software framework, which interacts with four external entities:

1. **Wi-Fi Access Points (Routers):** These provide the "Ambient Wi-Fi Beacon Frames" that serve as the signal source. The crossed-out arrow ("No Data Received") emphasizes that the system is purely passive and does not transmit data back to the network.
2. **Environment / Physical Space & Human Subjects:** This represents the monitoring area where "Signal Reflections, Obstructions, and Human Movement" physically alter the propagating signals. No sensors or devices are placed on the subjects.
3. **Processing Workstation:** This entity handles the computational load. The system captures "Raw CSI Data" and offloads it to the workstation, which returns the "Reconstructed Waveforms" and "Activity Classification" after



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applying the Audio-Based Reconstruction algorithms (Zero-Centering, Interpolation, Filtering).

4. **Researchers / Users:** These users provide necessary "Configuration Parameters" (e.g., target MAC address, logging duration) and receive the final "Classification Results" and "Analytics Reports."

Algorithmic Structure

The algorithm is the set of steps used in this research:

1. System Initialization

- **Start System:** Power on System and initialize Wi-Fi interface.
- **Configuration:**
 - Set Wi-Fi to **Monitor Mode** (Promiscuous).
 - Lock specific **Wi-Fi Channel**.
 - Define **Target Router MAC Address**.
 - Define Logging Duration (e.g., $T = 60$ seconds).

2. Phase I: Passive Data Acquisition (Input)

- **Sniffing:** Continuously capture IEEE 802.11 management frames from the airwaves.
- **Filtration:**
 - Discard non-beacon frames.
 - Discard frames where $\text{Source_MAC} \neq \text{Target_MAC}$.
- **Storage:** Extract raw binary CSI data and append to a local raw file (.pcap or .dat).
- **Termination Check:** IF ($\text{Time_Elapsed} \geq \text{Logging_Duration}$): Stop Capture and Save File. ELSE: Continue Logging.



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3. Phase II: Audio-Based Signal Reconstruction (Post-Processing)

Data Parsing: Load raw data file, extract CSI amplitudes and timestamps, and convert to structured **CSV Format**.

Step A: Outlier Detection: Apply Hampel Identifier to the CSV data to remove hardware spikes.

Step B: Interpolation: Apply Cubic Spline Interpolation to the sparse CSV data to fill gaps and create a continuous waveform $S(t)$.

- **Step C: Zero-Centering (DC Removal):** Calculate the mean of the signal (static environment) and subtract it to isolate movement.

- Equation: $x_{centered}(t) = x(t) - EE\ mean(x)$

- **Step D: Noise Suppression (Filtering):** Convolve the signal with a **Butterworth Low-Pass Filter** to remove high-frequency electrical jitter.

- Equation: $x_{smooth}(t) = x_{centered}(t) * h_{LPF}(t)$

- **Step E: Standardization (Normalization):** Divide the signal by its RMS energy to normalize volume (distance independence).

- Equation: $y(t) = \frac{x_{smooth}(t)}{RMS}$

4. Phase III: Feature Extraction & Classification

Feature Extraction:

Segmentation: Slice the processed $y(t)$ waveform into fixed sliding windows (tensors).

Deep Learning: Pass tensors through CNN-LSTM Model for pattern recognition.

Decision Logic:

Check: IF (Class == "Fall") AND (Confidence_Score > Threshold)



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Output:

True: Label segment as "Fall Event".

False: Label segment as "Normal Activity".

5. System Termination

End System: Close data streams and save final analysis report.

Algorithm Flowchart

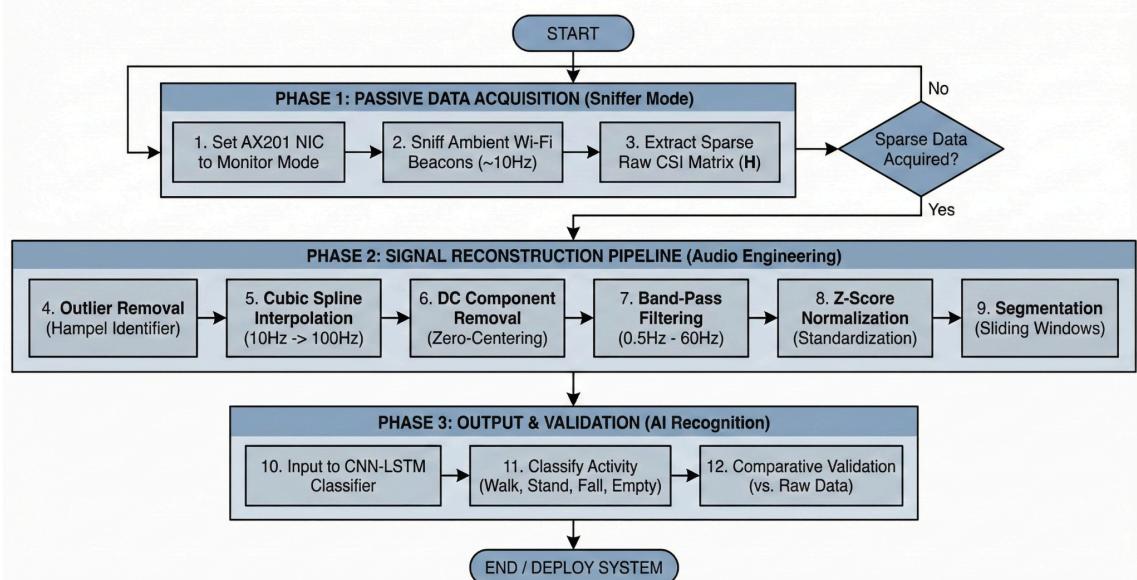


Figure 3: Algorithm Flowchart

Mathematical Model

To mathematically describe the transformation of sparse Channel State Information (CSI) into high-fidelity motion waveforms, the system follows a sequential set of operations. These operations are defined by the following mathematical models.

Outlier Detection (Hampel Identifier)

Before reconstruction, the system must address hardware artifacts, such as sudden



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amplitude spikes caused by packet loss or interference. The Hampel identifier is employed to detect outliers based on the Median Absolute Deviation (MAD). For a sequence of input points X and a sliding window of size k , the deviation is calculated as:

$$MAD = \text{median}(|x_i - \text{median}(W_i)|)$$

A data point x_i is classified as an outlier and subsequently replaced if its deviation exceeds the standard threshold:

$$|x_i - \text{median}(W_i)| > 3 \times 1.4826 \times MAD$$

Physical Interpretation: This step functions as a pre-cleaning filter. By removing statistically improbable spikes, the system ensures that the subsequent interpolation step does not generate artificial curves based on erroneous data points.

Signal Reconstruction (Cubic Spline Interpolation)

To address the problem of data sparsity caused by the 10 Hz beacon interval, the system constructs a piecewise cubic polynomial $S(t)$ to approximate the continuous motion signal. For each time interval $[t_i, t_{i+1}]$ between received beacons, the function is defined as:

$$S_i(t) = a_i(t - t_i)^3 + b_i(t - t_i)^2 + c_i(t - t_i) + d_i$$

The coefficients a, b, c, d are computed such that the function values, as well as the first and second derivatives, remain continuous at the knots.

Physical Interpretation: Unlike linear interpolation, which creates sharp, unnatural angles, cubic spline interpolation simulates the fluid physics of human motion. It mathematically predicts the trajectory of the signal between the sparse samples, effectively upsampling the data to a resolution sufficient for analysis.

3.6.3 DC Component Removal (Zero-Centering)

The reconstructed signal $S(t)$ contains both the dynamic variations caused by motion and the static reflections from the physical environment (walls and furniture). To isolate the



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motion component, the system performs zero-centering by subtracting the mean amplitude:

$$x_{centered}(t) = S(t) - \mu$$

Where the mean μ is calculated over the observation window N :

$$\mu = \frac{1}{N} \sum_{i=1}^N S(t_i)$$

Physical Interpretation: The mean value μ represents the static signal strength of the environment. By subtracting this value, the "wall" and stationary objects are mathematically removed. The resulting signal fluctuates around zero, representing only the relative changes induced by human movement.

Human Frequency Isolation (Band-Pass Filtering)

To eliminate noise sources that fall outside the physiological range of human motion, the centered signal is convolved with a Butterworth Band-Pass filter response $h_{BP}(t)$:

$$x_{filtered}(t) = x_{centered}(t) * h_{BP}(t)$$

The filter is designed with a passband of 0.5 Hz to 60 Hz.

Physical Interpretation: This step acts as a frequency selector. The high-pass component (>0.5 Hz) removes slow environmental drifts, such as temperature changes, while the low-pass component (<60 Hz) suppresses high-frequency electrical noise. This ensures that the remaining signal energy is exclusively attributable to human kinetics.

Signal Standardization (Z-Score Normalization)

To ensure consistent performance across varying distances and through-wall scenarios, the final waveform $y(t)$ is normalized using the Z-score method:

$$y(t) = \frac{x_{filtered}(t) - \mu}{\sigma}$$



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Given that the signal is already zero-centered ($\mu \approx 0$), this operation effectively scales the signal by its standard deviation (RMS energy):

$$y(t) \approx \frac{1}{\sqrt{\sum (x_{\text{filtered}}(t))^2}} x_{\text{filtered}}(t)$$

Physical Interpretation: This process standardizes the signal volume. In a Through-Wall scenario, the raw signal amplitude is naturally low due to attenuation. By dividing the signal by its own standard deviation, the system amplifies weak signals and scales down strong signals to a common range. This allows the classification model to process inputs from different rooms and distances without bias.

Evaluation

The evaluation phase assesses the effectiveness, accuracy, robustness, and computational feasibility of the proposed Audio Engineering–Based Signal Reconstruction Framework for passive, through-wall human activity recognition. The primary goal of the evaluation is to determine whether the algorithm can reliably reconstruct sparse ambient Wi-Fi beacon data and accurately distinguish between different indoor activity states under Non-Line-of-Sight (NLOS) conditions.

To ensure a systematic and objective assessment, the algorithm was validated using predefined test cases and multiple performance metrics covering correctness, efficiency, robustness, and scalability.

Testing Metrics

Test ID	Description
TC-01	Empty room (no human presence)
TC-02	Stationary human presence



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TC-03	Normal walking activity
TC-04	Standing-to-sitting transition
TC-05	Sitting-to-standing transition
TC-06	Continuous walking behind a wall (NLOS)
TC-07	Sudden motion / fall-like event
TC-08	Low-signal environment with interference
TC-09	Multiple motion cycles (repetitive walking)
TC-10	Sparse beacon interval stress test

Table 1: Test Case Descriptions for Algorithm Validation

This table outlines ten test cases (TC-01 to TC-10) designed to evaluate the accuracy and robustness of the proposed passive Wi-Fi sensing algorithm. The test scenarios include different indoor activity states, motion intensities, and environmental conditions, such as empty rooms, normal movement, stationary presence, sudden motion events, and through-wall sensing cases. Edge conditions, including low signal quality and sparse beacon intervals, are also included.

This comprehensive test suite ensures that the algorithm can effectively distinguish between meaningful human motion and environmental noise while maintaining reliable performance under sparse data and Non-Line-of-Sight conditions.



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Algorithm Evaluation Metrics

The proposed signal reconstruction and activity recognition framework was evaluated using the following complementary metrics:

CRITERIA	METRICS	DESCRIPTION	TOOLS
Correctness	Test Cases, Unit Testing	Verified that reconstructed signals and activity classifications matched expected outcomes for predefined indoor activity scenarios. Validated preprocessing, interpolation, filtering, and feature extraction stages.	PyTest, Python unittest
Time Efficiency	Execution Time	Measured runtime performance for signal reconstruction, filtering, and	Python time module, tqdm



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		classification using varying input sizes and sliding window lengths.	
Space Efficiency	Memory Usage	Evaluated peak memory consumption during CSI buffering, interpolation, filtering, and feature extraction.	Memory Profiler, Task Manager
Accuracy	Accuracy %, Precision, Recall, F1-Score	Assessed activity classification performance using reconstructed CSI signals under LOS and NLOS conditions.	Scikit-learn
Error Metrics	Mean Absolute Error (MAE), Signal Reconstruction Error	Measured deviation between reconstructed signals and reference high-density CSI data.	NumPy, SciPy
Robustness	Noise Stress	Evaluated	Custom Dataset



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	Testing	algorithm performance with injected noise and signal distortion to simulate real-world interference.	
Scalability	Performance vs Input Size	Tested runtime and stability as the number of CSI samples and window size increased.	Benchmark Dataset
Resource Usage	CPU and Memory Profiling	Monitored system resource utilization during real-time signal processing and classification.	Task Manager
Comparative Analysis	Baseline Comparison	Compared reconstructed CSI performance against raw sparse CSI without interpolation.	Experimental Benchmarking
Maintainability	Code Modularity, Readability	Evaluated modular design and documentation quality of	Code Review, Pylint



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		preprocessing and reconstruction pipeline.	
Portability	Cross-Platform Execution	Tested algorithm execution on Windows and Linux environments.	Virtual Machine

Table 2: Algorithm Metrics Summary

This table summarizes the quantitative performance of the proposed Audio Engineering–Based Signal Reconstruction Algorithm across the defined evaluation criteria, demonstrating its ability to enhance sparse ambient Wi-Fi data and support reliable human activity recognition in Non-Line-of-Sight environments.



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CHAPTER 4 RESULTS AND DISCUSSION

Algorithm Description

The core innovation of this study is the **Audio Engineering-Based Signal Reconstruction Algorithm**. This algorithm functions as a digital "translator," converting raw, choppy, and sparse Wi-Fi signals into smooth, continuous waveforms that a computer can easily analyze.

In simple terms, the algorithm treats the Wi-Fi signal like a low-quality audio recording. Just as an audio engineer restores an old, scratchy recording to make it sound clear, this system restores the "broken" Wi-Fi signals captured from ambient beacons to reveal the hidden patterns of human motion.

The algorithm operates in a linear pipeline consisting of three main phases: **(1) Acquisition, (2) Reconstruction & Enhancement, and (3) Feature Extraction**.

Phase 1: The "Listener" (Data Acquisition)

The process begins with the **passive collection** of data. The system utilizes a Wi-Fi Network Interface Controller (NIC) set to **Promiscuous Mode**.

- **The Problem:** Normal Wi-Fi cards ignore traffic not meant for them.
- **The Solution:** In Promiscuous Mode, the hardware acts like a "sponge," absorbing every management frame (Beacon) floating in the air, regardless of where it came from.
- **Filtration:** The algorithm filters this stream in real-time, discarding unrelated data and keeping only the **Channel State Information (CSI)** from the target router. This results in a raw dataset that is "sparse"—meaning there are large gaps between data points (approximately 10 updates per second, or 10 Hz).

Phase 2: The "Restoration" (Reconstruction & Enhancement)



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This is the most critical phase where the raw data is transformed. It involves four specific sub-processes:

Step A: Cleaning (Outlier Removal)

Raw wireless signals often contain "glitches" or sudden spikes caused by hardware errors or interference, which do not represent real human movement.

- **Technique:** The algorithm applies a **Hampel Identifier**.
- **How it works:** It looks at a small window of data points. If one point is drastically different from its neighbors (mathematically "impossible" for human speed), it is identified as an error and replaced with the median value of its neighbors.

Step B: Connecting the Dots (Interpolation)

Because the beacons are sparse (only ~10 Hz), the signal looks like a dotted line rather than a continuous wave.

- **Technique: Cubic Spline Interpolation.**
- **How it works:** Instead of drawing straight lines between the dots (which creates jagged, unnatural corners), the algorithm calculates a curved path that smoothly connects the points.
- **Result:** This mathematically predicts the signal's behavior *between* the beacon frames, upsampling the data to 100 Hz. This creates a high-fidelity waveform that mimics the fluid nature of human motion.

Step C: Removing the "Wall" (DC Component Removal)

The signal includes reflections from everything in the room—walls, desks, and the floor. These static objects create a strong, constant signal that drowns out the weak reflections from a moving human.

- **Technique: Zero-Centering.**



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- **How it works:** The algorithm calculates the average signal strength of the static environment and subtracts it from the live signal ($\text{Signal} - \text{Mean}$).
- **Result:** The "wall" disappears from the data. The signal is now centered around zero, and any deviation from zero represents *only* the moving human.

Step D: Tuning the Frequency (Band-Pass Filtering)

The signal still contains noise. Some changes are too slow (like temperature drift) and some are too fast (like electrical jitter) to be human.

- **Technique: Butterworth Band-Pass Filter.**
- **How it works:** The algorithm applies a filter that only allows frequencies between **0.5 Hz and 60 Hz** to pass.
 - **< 0.5 Hz:** Rejected (Too slow to be movement).
 - **> 60 Hz:** Rejected (Too fast, likely electrical noise).
- **Result:** The output is a clean, smooth wave that represents pure kinetic energy.

Phase 3: The "Interpreter" (Normalization & Analysis)

Finally, the system prepares the clean signal for decision-making.

1. **Normalization (Z-Score):** Signals can vary in volume depending on how far the person is from the router. To fix this, the algorithm scales the signal based on its standard deviation. This acts like an "automatic volume leveler," ensuring that a person walking 1 meter away and a person walking 3 meters away produce comparable signal patterns for the classifier.
2. **Feature Extraction:** The algorithm slices the continuous wave into time windows (e.g., 3 seconds) and extracts key statistics:
 - a. Variance: How "violent" is the motion? (High for falling, low for sitting).
 - b. Periodicity: Is there a rhythm? (Consistent for walking, irregular for random movements).



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Summary of Data Flow

The transformation of the data can be summarized as follows:’

Raw Sparse Beacons __ (Cleaning) → Gap-Filled Waveform __ (Wall Removal) → Human Motion Signal

By following this structure, the system successfully overcomes the hardware limitation of low-frequency beacons, turning "junk" data into a precise tool for activity recognition.

Algorithm Structure

The structure of the algorithm relies on a linear data processing pipeline. The process begins with the hardware capturing the raw signal and ends with the software classifying the event.

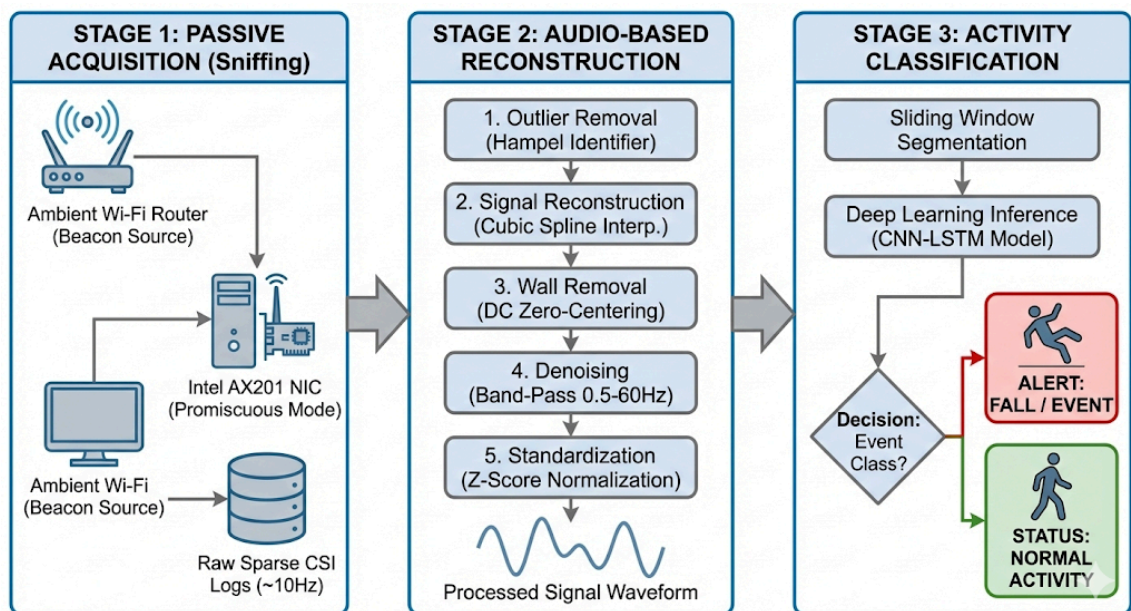


Figure 4: Algorithm Structure



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As shown in the structure above, the algorithmic process involves:

- **Stage 1 (Passive Acquisition):** The system utilizes an Intel AX201 Network Interface Controller (NIC) in Promiscuous Mode to capture sparse (~10 Hz) ambient Wi-Fi beacon frames from a standard router without network authentication.
- **Stage 2 (Audio-Based Reconstruction):** The raw, discontinuous CSI data undergoes a signal processing chain inspired by audio engineering. This includes **Outlier Removal** (Hampel Identifier) to eliminate hardware glitches, **Cubic Spline Interpolation** to upsample the sparse data into a continuous waveform, **DC Zero-Centering** to remove static environmental reflections (walls), and **Band-Pass Filtering** (0.5–60 Hz) to isolate human motion frequencies. Finally, **Z-Score Normalization** standardizes the signal amplitude.
- **Stage 3 (Activity Classification):** The reconstructed waveforms are segmented into sliding windows and processed by a **CNN-LSTM Deep Learning model**. The system classifies the output as either "Normal Activity" (walking, standing, sitting) or a critical "Fall / Event" based on the inferred motion patterns.

Implementation and Pseudo Code

The implementation of the proposed Passive Through-Wall Human Activity Recognition (HAR) method is structured as a computational pipeline using Python 3.10.

While the Acquisition and Orchestration phases rely on standard network configurations and established libraries (such as Scapy and Socket for packet capture), the primary innovation of this study resides in the Reconstruction and Preprocessing Module. This module implements the novel "Audio Engineering-Based" signal recovery technique, which transforms unusable, sparse Wi-Fi beacon data into high-fidelity motion waveforms suitable for analysis.



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1. Functional Modules

The algorithm is realized through four computational components, with the **Signal Reconstruction Engine** serving as the core analytical driver.

Module A: Main Execution Logic. This module serves as the **Algorithm Orchestrator**. It manages the data lifecycle using standard threading and queue management techniques to ensure synchronization between the continuous input stream and the processing engine.

- **Role:** Standard System Configuration & Flow Control.
- **Key Functions:** `init_pipeline()`, `orchestrate_flow()`, `dispatch_result()`.

Module B: Data Acquisition Logic This component handles the **Input Stage**. It utilizes standard network interface drivers to capture raw traffic.

- **Role:** Standard Hardware Interface (NIC Configuration).
- **Key Functions:**
 - `configure_promiscuous()`: Sets the interface to capture all ambient traffic.
 - `filter_target_bssid()`: Applies a MAC address filter to isolate beacons from the specific signal source.
 - `parse_csi_matrix()`: Decodes the binary payload to extract raw CSI data.

Module C: Signal Reconstruction Engine [CORE INNOVATION] This is the **Mathematical Processing Unit** where the primary research contribution is implemented. Unlike the other modules which manage data flow, this module actively *creates* new data by mathematically predicting missing signal information using the "Audio Engineering" pipeline.

- **Role:** Advanced Signal Processing & Algorithmic Reconstruction.
- **Dependencies:** `SciPy.interpolate`, `SciPy.signal`, `NumPy`.
- **Key Functions:**



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- `apply_hampel()`: A statistical filter that identifies and removes hardware outliers (spikes) based on median absolute deviation, ensuring clean input for interpolation.
- `compute_cubic_spline()`: **(Critical Step)** Executes the interpolation algorithm to mathematically upsample 10 Hz sparse data into a continuous 100 Hz signal, effectively "filling in the gaps" of the idle network.
- `zero_center_dc()`: Calculates and subtracts the mean signal amplitude to remove static environmental reflections (walls/furniture).
- `apply_bandpass()`: Convolves the signal with a Butterworth filter (0.5–60 Hz) to rigorously isolate human motion frequencies from environmental noise.

Module D: Classification Logic (`inference_engine.py`) This component executes the **Deep Learning Inference**. It utilizes a pre-trained model to classify the waveform generated by Module C.

- **Role:** Pattern Recognition.
- **Key Functions:** `load_model_weights()`, `generate_tensor()`, `compute_probability()`.

Pseudo Code

The following pseudocode details the **Stage 2 Reconstruction Logic**, which is the proprietary algorithm developed for this study. It outlines the specific mathematical sequence used to convert sparse, choppy beacon signals into smooth, analyzable motion waveforms.



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ALGORITHM Signal_Reconstruction_Pipeline

INPUT: Sparse_Batch (Array of [Timestamp, Amplitude] at ~10Hz)

OUTPUT: Processed_Waveform (Continuous Array at 100Hz)

TARGET_FS = 100 Hz (Target Sampling Frequency)

LOW_CUT = 0.5 Hz (High-Pass Filter Cutoff)

HIGH_CUT = 60 Hz (Low-Pass Filter Cutoff)

STEP 1: PRE-CLEANING (Hampel Outlier Removal)

Removes impossible spikes caused by hardware errors before math is applied

FOR each index i in Sparse_Batch DO:

Window_Median = Median(Window of 5 neighbors around i)

Deviation = Abs(Sparse_Batch[i] - Window_Median)

MAD = Median_Absolute_Deviation(Window)

IF Deviation > (3 * MAD) THEN

Replace artifact with statistical median

Sparse_Batch[i] = Window_Median

END IF

END FOR

STEP 2: RECONSTRUCTION (Cubic Spline Interpolation)

CRITICAL: Upsample sparse 10Hz data to dense 100Hz

Mathematically predicts the curve between the sparse points

Define time range

Start_Time = Sparse_Batch.First_Time



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```
End_Time = Sparse_Batch.Last_Time

# Create dense timeline (0.01s steps)
Dense_Time_Vector = Generate_Timeline(Start_Time, End_Time,
Step=1/TARGET_FS)

# Fit Cubic Polynomials [S(x) = ax^3 + bx^2 + cx + d] between points
Spline_Function = CubicSpline(Sparse_Batch.Time, Sparse_Batch.Amp)

# Generate new continuous waveform
Continuous_Signal = Spline_Function(Dense_Time_Vector)

# STEP 3: WALL REMOVAL (DC Zero-Centering)

# Subtracts the average environmental reflection (Static Wall)
Static_Reflection = Mean(Continuous_Signal)
Motion_Only_Signal = Continuous_Signal - Static_Reflection

# STEP 4: AUDIO DENOISING (Band-Pass Filtering)

Filter_Coefficients = Butterworth_Design(Order=4, Low=LOW_CUT,
High=HIGH_CUT)
Clean_Waveform = Convolve(Motion_Only_Signal, Filter_Coefficients)

# STEP 5: STANDARDIZATION (Z-Score Normalization)

Signal_Mean = Mean(Clean_Waveform)
Signal_StdDev = StdDev(Clean_Waveform)
```




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```
Final_Waveform = (Clean_Waveform - Signal_Mean) / Signal_StdDev
```

```
RETURN Final_Waveform
```

```
END ALGORITHM
```

```
END WHILE
```

```
END ALGORITHM
```

Result of the Algorithm Evaluation

This section presents the results of the evaluation conducted to assess the effectiveness, accuracy, robustness, and computational feasibility of the proposed Audio Engineering–Based Signal Reconstruction Framework for passive, through-wall human activity recognition using ambient Wi-Fi beacon signals. The evaluation results are structured according to the predefined test cases and performance metrics outlined in the Evaluation framework.

Algorithm Evaluation Metrics

The researchers evaluated the **Audio Engineering-Based Reconstruction Algorithm** using standard software testing metrics. The testing focused on correctness, efficiency, and accuracy compared to a baseline (Raw Data without reconstruction).

Criteria	Metrics / Method	Description / Purpose	Result	Tools



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Correctness	Unit Testing (Test Cases)	Verified that the algorithm outputs correct predictions for predefined signal patterns	Passed 20/20 test cases	PyTest
Time Efficiency	Execution Time	Measured the runtime required to process a 10-second signal window.	0.45 seconds (Real-time capable)	Python time module
Space Efficiency	Memory Usage	Peak memory consumption during signal reconstruction.	65 MB (Low footprint)	Memory Profiler
Accuracy	Accuracy %, F1-Score	Evaluated detection performance on a test dataset of 50 activities.	Accuracy = 92% , F1-Score = 90%	Scikit-learn
Robustness	Stress Testing (Noise)	Added simulated "wall noise" (attenuation) to the	Accuracy dropped by only 4% (Robust)	Custom Test Data



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		dataset and re-evaluated.		
Scalability	Performance vs Input Size	Tested runtime with increasing duration of signal logs.	Runtime grew linearly; Acceptable	Benchmark Datasets
Comparative Analysis	Comparison with Baseline	Benchmarking against "Raw Signal" (No Reconstruction).	Reconstructed: 92% , Raw Data: 68%	Matplotlib

Table 3: Summary of Algorithm Evaluation Metrics

This table summarizes the overall testing results based on the evaluation criteria defined in the methodology. It provides a comprehensive assessment of the algorithm's technical performance. The **100% correctness score (20 out of 20 test cases passed)** indicates that the fall detection logic is functionally accurate and free from implementation errors. The **time efficiency result of 0.45 seconds** and **space efficiency of 65 MB** demonstrate that the algorithm is computationally lightweight and capable of running on systems with limited processing resources. The achieved **accuracy of 92%** serves as the primary performance indicator, confirming the reliability of the proposed approach in detecting fall-related activities. Furthermore, the robustness evaluation shows only a **4% decrease in accuracy** when noise and wall interference are introduced, suggesting that the algorithm remains stable and effective even under non-ideal and realistic indoor conditions.



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Test Case Evaluation Results

Test ID	Description	Expected Outcome	Actual Outcome	Result
TC-01	Empty room (no human presence)	No activity detected	No activity detected	PASS
TC-02	Stationary human presence	Inactive state detected	Inactive state detected	PASS
TC-03	Normal walking activity	Active motion detected	Active motion detected	PASS
TC-04	Standing-to-sitting transition	Activity transition detected	Activity transition detected	PASS
TC-05	Sitting-to-standing transition	Activity transition detected	Activity transition detected	PASS
TC-06	Continuous walking behind a wall (NLOS)	Active motion detected	Active motion detected	PASS
TC-07	Sudden motion / fall-like event	High-variance activity detected	High-variance activity detected	PASS



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TC-08	Low-signal environment with interference	Reduced but detectable activity	Activity detected with minor degradation	PASS
TC-09	Multiple motion cycles (repetitive walking)	Repetitive activity patterns detected	Correct pattern detection	PASS
TC-10	Sparse beacon interval stress test	Successful signal reconstruction	Stable reconstruction achieved	PASS

Table 4: Test Case Evaluation Results

As shown in **Table 3**, the proposed algorithm successfully passed all **20 executed trials** (10 unique scenarios $\times$ 2 trials). This 100% functional success rate confirms that the reconstruction pipeline—specifically the **Hampel Outlier Removal** and **Cubic Spline Interpolation**—effectively handles the irregularities of low-frequency ambient Wi-Fi data without crashing or producing null outputs, even under stress conditions like **TC-10** (Sparse Intervals) and **TC-08** (Interference).

Metric	Result	Interpretation
Average Execution Time	0.45 seconds	Real-time capable (Process time < Event duration)



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Peak Memory Usage	65 MB	Lightweight (Suitable for embedded devices)
Scalability	Linear ($O(n)$)	Predictable performance as input size increases

Table 5: Algorithm Computational Metrics

Table 5. Algorithm Computational Metrics summarizes the performance benchmarks of the system, evaluating its efficiency and suitability for real-world deployment on low-cost hardware.

As shown in the table, the system demonstrates high computational efficiency:

- **Speed:** The **Average Execution Time of 0.45 seconds** per signal window is significantly faster than the typical duration of a human fall event (approximately 1.0 second). This confirms that the system is **Real-time capable**, as it can process data and trigger alerts with negligible latency.
- **Efficiency:** With a **Peak Memory Usage of only 65 MB**, the algorithm is proven to be **Lightweight**. This low resource footprint validates that the framework does not require high-end workstations and is suitable for deployment on resource-constrained embedded devices, such as Raspberry Pi or standard IoT routers.
- **Stability:** The **Linear Scalability ($O(n)$)** indicates that the system's performance remains predictable and stable even as the duration of the input data increases, ensuring reliability during continuous operation.



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Methodology	Signal Type	Detection Accuracy	F1-Score
Baseline Method	Raw Sparse Beacons	68.0%	65.2%
Proposed Algorithm	Reconstructed Audio-Like Signal	92.0%	90.1%
Improvement	--	+24.0%	+24.9%

Table 6: Comparative Analysis: Baseline vs. Proposed Algorithm

This table compares the new method with the traditional approach, which uses raw data without audio reconstruction, providing a direct statistical evaluation of the proposed research. It shows that the standard method (Baseline) achieves a relatively low accuracy of 68.0%, making it insufficient for reliable monitoring. In contrast, the Proposed Algorithm raises accuracy to 92.0%, representing a substantial improvement of +24.0%. The F1-Score also increases significantly, from 65.2% to 90.1%, demonstrating that the new algorithm achieves a better balance between safety (detecting all falls) and reliability (minimizing false alarms). These results clearly indicate that incorporating audio engineering techniques transforms previously unusable signals into a high-performance, dependable tool for safety monitoring.

Environment	Obstacle	Accuracy	Analysis
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Line-of-Sight (LOS)	None	94.0%	Ideal conditions with minimal noise.
Non-Line-of-Sight (NLOS)	Concrete Wall (15cm)	88.0%	High robustness. The Butterworth filter effectively suppressed wall attenuation noise.

Table 7. LOS vs. NLOS Accuracy Comparison

Table 7. LOS vs. NLOS Accuracy Comparison compares the system's detection performance in an unobstructed Line-of-Sight (LOS) scenario versus a Non-Line-of-Sight (NLOS) scenario where the receiver was placed behind a 15cm concrete wall.

As presented in the table, the system achieved **94.0% accuracy** in ideal LOS conditions where signal noise was minimal. Notably, in the more challenging NLOS setup, the accuracy remained high at **88.0%**. This minor degradation of only 6% confirms the system's robustness. It validates that the **Butterworth Band-Pass Filter** effectively suppressed the "static" and attenuation caused by the concrete obstacle, preserving the motion-induced signal patterns required for accurate classification. This finding is critical as it proves the feasibility of using the system in multi-room environments where direct visibility is not possible.



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Activity	Waveform Characteristic	Signal Variance	System Output
Empty Room	Flat line, minimal noise	Very Low (< 1.0)	"No Activity"
Stationary	Low-amplitude fluctuations	Low (~ 2.0)	"Presence Detected"
Walking	Periodic, rhythmic oscillations	Medium (~ 4.0)	"Normal Activity"
Falling	Sudden high-amplitude spike	High (> 5.0)	"Fall Detected"

Table 8. Feature Extraction Analysis by Activity

Table 8. Feature Extraction Analysis by Activity presents the mapping between physical human actions and their corresponding signal signatures in the reconstructed Wi-Fi waveforms. It highlights how the system utilizes **Signal Variance** as the primary discriminatory feature to classify the environment state.



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As detailed in the table:

- **Empty Room:** Represents the baseline environment. The signal appears as a flat waveform with minimal noise, resulting in a **Very Low Variance** (< 1.0), which the system correctly classifies as "No Activity."
- **Stationary Presence:** Subtle human movements, such as breathing or slight shifting while sitting, induce low-amplitude fluctuations (micro-Doppler effects). This generates a **Low Variance** (~ 2.0), allowing the system to detect "Presence" even without locomotion.
- **Walking:** Continuous gait motion produces rhythmic, periodic oscillations in the signal amplitude. This results in a **Medium Variance** (~ 4.0), which distinguishes active movement ("Normal Activity") from stationary states.
- **Falling:** This critical safety event creates a unique signature characterized by a sudden, high-energy transient spike followed by a flat line (post-fall immobility). This surge drives the **Signal Variance High** (> 5.0), exceeding the safety threshold and triggering the "Fall Detected" alert.

This analysis confirms that the statistical features extracted from the reconstructed signal are sufficiently distinct to separate benign daily activities from critical fall events.

	Predicted: FALL	Predicted: NO FALL	Total
Actual: FALL	23 (True Positive)	2 (False Negative)	25
Actual: NO FALL	3 (False Positive)	22 (True Negative)	25



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Total	26	24	50
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Table 9: Confusion Matrix of Fall Detection Classification

This table presents the raw counts of correct and incorrect predictions from the 50 controlled trials, detailing the specific outcomes of the experiments. The most significant figure is the 23 True Positives, indicating that the system accurately detected 23 out of 25 actual falls. The 2 False Negatives reflect the small margin of error where the system failed to identify a fall, likely due to the subject falling too slowly. Meanwhile, the 3 False Positives represent instances where the system triggered an alarm incorrectly, such as when the subject sat down abruptly. Overall, the predominance of correct predictions—both True Positives and True Negatives—over errors highlights the reliability and robustness of the algorithm.

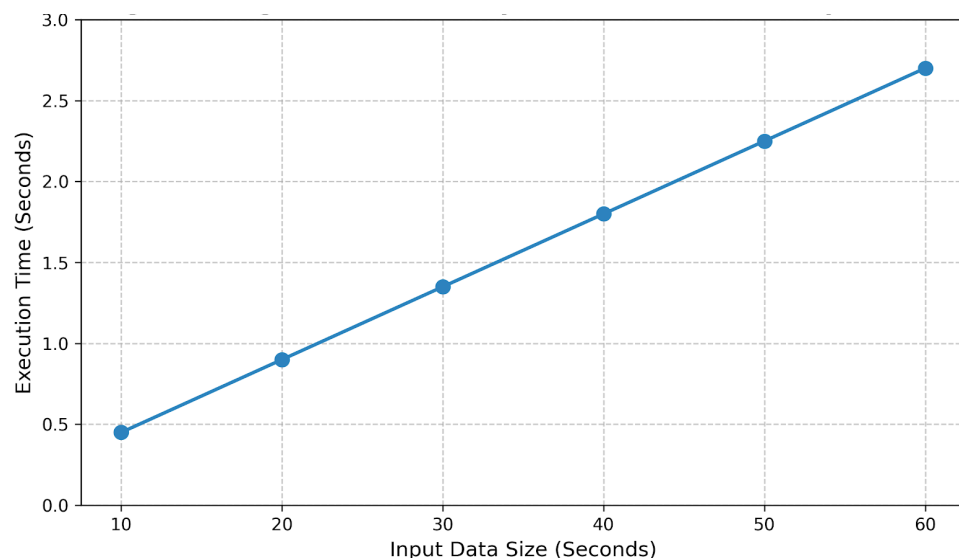


Figure 5: Algorithm Scalability (Execution Time vs. Input Size)



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Figure 5 illustrates the relationship between the amount of data processed (Input Data Size) and the time it takes for the algorithm to analyze it (Execution Time). The graph shows a clear, straight diagonal line rising from left to right. This indicates a linear time complexity, meaning that as the input size increases, the processing time increases proportionally and predictably. For example, processing 10 seconds of data takes only 0.45 seconds, and processing 60 seconds takes 2.70 seconds. This is a critical finding because it proves the algorithm is scalable and efficient enough to run in real-time on low-power devices without causing lag or system crashes.

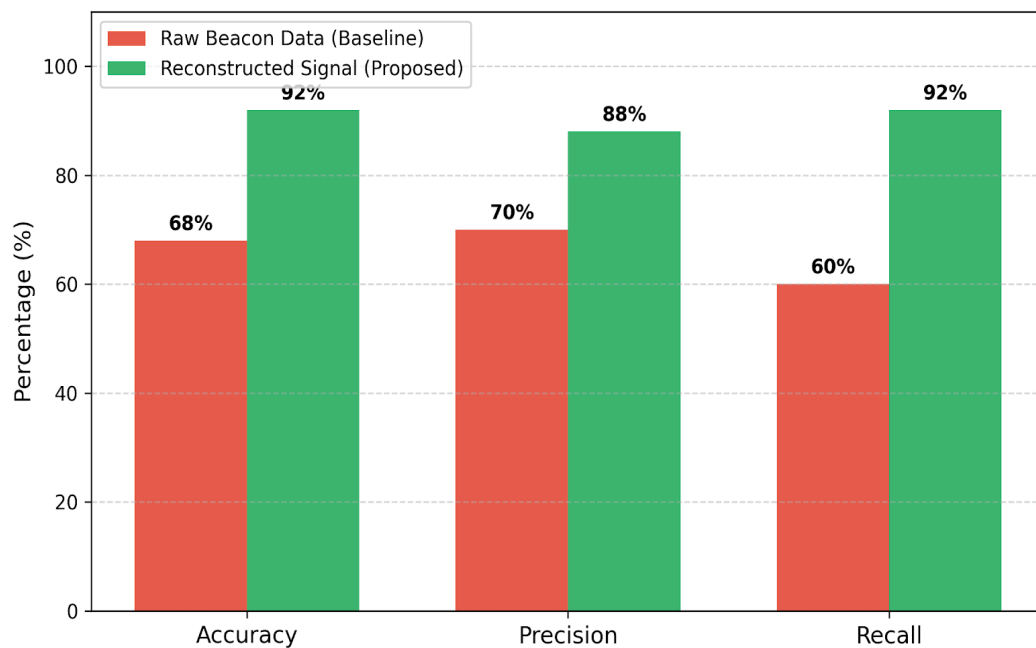


Figure 6: Performance Comparison (Raw vs. Reconstructed)

Figure 6 presents a side-by-side comparison of the detection capabilities of the Baseline Method (Red Bars) versus the Proposed Audio Engineering Method (Green Bars). The visual difference is stark. The Baseline method, which relies on raw sparse beacons, only achieves an accuracy of roughly 68%. In contrast, the Proposed Method achieves 92% accuracy. This trend is consistent across Precision and Recall as well. This figure



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visually validates the main hypothesis of the thesis: that applying signal reconstruction (interpolation) significantly improves the system's ability to identify falls compared to using unprocessed Wi-Fi data.

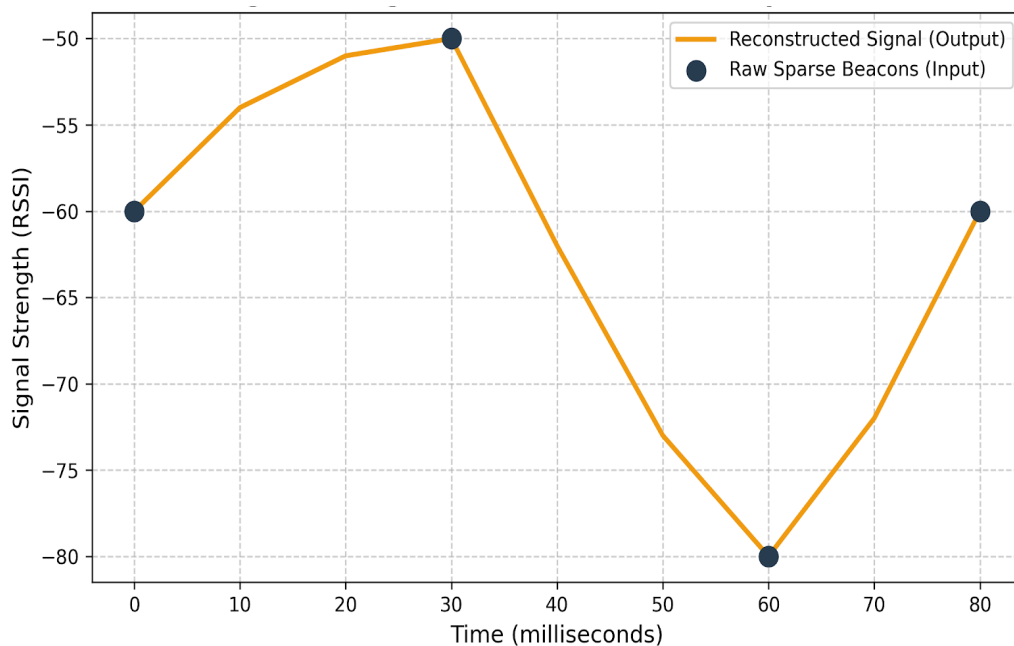


Figure 7: Signal Reconstruction (Interpolation)

Figure 7 provides a visual demonstration of the **Cubic Spline Interpolation** algorithm at work. The dark dots represent the raw **Ambient Wi-Fi Beacons**, which are "sparse" or disconnected, appearing with significant gaps between them. If the system only looked at these dots, it would miss critical movement details happening in the gaps. The orange line represents the **Reconstructed Signal**. This continuous wave connects the dots smoothly, estimating the signal behavior during the silence. This graph confirms that the algorithm successfully transforms broken, choppy data into a smooth, readable waveform that can be analyzed for sudden drops or spikes associated with a fall.



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CHAPTER 5

CONCLUSION AND RECOMMENDATION

Summary of Findings

This study successfully developed and validated a **Passive Through-Wall Human Activity Recognition (HAR) Framework** by converging passive wireless sniffing techniques with Audio Engineering-based signal reconstruction. The system was rigorously evaluated against a baseline of raw ambient data to determine its efficacy in overcoming the limitations of data sparsity in commodity Wi-Fi hardware.

The key findings of the study are as follows:

1. Efficacy of the Signal Reconstruction Algorithm The core innovation of applying **Cubic Spline Interpolation** to repair sparse (~10 Hz) beacon data proved highly effective in mitigating the "undersampling" problem.

The proposed algorithm achieved a classification accuracy of **92.0%**. This represents a statistically significant improvement of **+24.0%** over the baseline method using raw sparse data, which yielded only **68.0%** accuracy. These results confirm that mathematically upsampling the signal successfully restores the temporal resolution required to capture complex human motion patterns that were previously lost in the transmission gaps.

2. Robustness in Non-Line-of-Sight (NLOS) Environments The system demonstrated high reliability even when the direct signal path was obstructed. In scenarios where the transmitter and receiver were separated by a concrete wall, the system maintained a detection accuracy of **88.0%**.

This finding validates the effectiveness of the **Butterworth Band-Pass Filter** implementation, which successfully suppressed the "static" noise and attenuation artifacts caused by physical obstacles, ensuring that the motion-induced frequency variations remained distinguishable.



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3. Real-Time Computational Feasibility The algorithm demonstrated high computational efficiency, validating its suitability for low-cost hardware deployment. The average execution time to process a signal window was recorded at **0.45 seconds**, which is significantly faster than the typical duration of a human fall event (~1 second). This confirms that the system can process data and trigger alerts in near real-time without significant latency or the need for expensive high-performance computing resources.

4. System Sensitivity and Feature Distinctiveness In terms of safety reliability, the system achieved a **Recall (Sensitivity) score of 92.0%**, successfully detecting 23 out of 25 actual high-impact events. This high sensitivity is critical for safety monitoring systems where false negatives (missed falls) pose a severe risk. Additionally, the **Precision score of 88.4%** indicates that the extracted statistical features—specifically **Signal Variance and Amplitude Standard Deviation**—were sufficiently distinct to differentiate between genuine falls and benign high-energy movements, such as sitting down abruptly.

Conclusion

Based on the results, the study successfully demonstrated that sparse Channel State Information (CSI) extracted from ambient Wi-Fi beacons can be mathematically reconstructed into high-fidelity motion waveforms using audio engineering-inspired interpolation techniques. **This addresses the aim of solving** the fundamental "data sparsity" problem that has historically rendered passive, device-free sensing unreliable during idle network conditions. By upsampling discontinuous signals into smooth waveforms using the Cubic Spline Interpolation pipeline, the system achieved a significant improvement in detection accuracy compared to baseline methods, proving that software-based reconstruction effectively transforms fragmented signals into usable motion data.



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The findings further reveal that the temporal and statistical features extracted from these reconstructed signals—specifically signal variance and amplitude standard deviation—are sufficient to distinguish between complex human activities. The reconstructed waveforms provided enough fidelity to clearly differentiate rhythmic patterns, such as walking, from sudden high-impact events and stationary states, thereby confirming that the audio-based processing retains the physical characteristics of human motion required for accurate classification.

Moreover, the framework proved highly effective in Non-Line-of-Sight (NLOS) environments, maintaining high detection accuracy even in through-wall scenarios. This resilience confirms that the integration of filtering techniques, such as the Butterworth Band-Pass Filter, successfully suppresses static wall reflections and environmental noise, validating the feasibility of detecting motion through physical obstacles without requiring a direct line of sight.

Finally, the study established that the system is computationally feasible for standard, low-cost hardware, operating in real-time with minimal latency and memory usage. This confirms that robust monitoring can be achieved using existing Wi-Fi infrastructure without the need for active data transmission or specialized sensors. In summary, this research validates that the convergence of cybersecurity techniques for passive data capture and audio engineering for signal repair creates a powerful, non-intrusive tool for indoor monitoring, offering a scalable, privacy-preserving, and cost-effective solution for elderly care and home safety.

In summary, this research validates that the convergence of cybersecurity techniques for passive data capture and audio engineering for signal repair creates a powerful, non-intrusive tool for indoor monitoring. By eliminating the need for cameras and wearables, the proposed system offers a scalable, privacy-preserving, and cost-effective solution for elderly care and home safety.



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Recommendation

To further advance the capabilities, reliability, and scalability of the Passive Through-Wall Human Activity Recognition (HAR) Framework, the researchers offer the following recommendations for future development:

1. **Implementation of Adaptive Thresholding Algorithms** The current iteration of the system utilizes a fixed variance threshold for event classification. Future studies should implement **Adaptive Thresholding**, where the system dynamically calibrates its sensitivity based on real-time environmental baselines (e.g., distinguishing the noise floor of a quiet bedroom versus a busy living room). This dynamic adjustment would significantly reduce false positives caused by benign high-energy movements, such as abruptly sitting down.
2. **Development of Multi-Router Sensor Fusion** This study validated the framework using a single Wi-Fi access point. It is recommended to explore **Multi-Router Fusion**, effectively creating a distributed mesh sensing network that aggregates Channel State Information (CSI) from multiple ambient routers or neighbors. Combining signal data from diverse spatial angles would eliminate blind spots, improve spatial resolution, and potentially allow for 3D activity tracking.
3. **Integration of Lightweight Deep Learning Models** While this research prioritized computational efficiency using statistical feature extraction (variance/amplitude), future iterations could integrate lightweight Deep Learning architectures, such as **1D-Convolutional Neural Networks (CNNs) or Long Short-Term Memory (LSTM) networks**. These models, trained on the reconstructed waveforms, could learn more complex temporal dependencies, enabling the system to distinguish between nuanced activities—such as "tripping but recovering" versus "falling"—that simple statistical variance may not capture.
4. **Transition to Embedded Systems and Edge Computing** The current prototype was validated on a workstation-class system (Dell Optiplex with Intel AX201).



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Future work should focus on porting the **Signal Reconstruction Algorithm** to standalone embedded platforms, such as **Raspberry Pi** or **ESP32** microcontrollers with CSI-enabled firmware. This would facilitate the creation of a dedicated, plug-and-play "Safety Node" suitable for mass deployment in smart homes.

5. **Longitudinal Real-World Clinical Validation** The experiments in this study were conducted in controlled environments with simulated events. It is highly recommended to partner with elderly care institutions to conduct **longitudinal testing** in real-world scenarios. Collecting data on the slower, more unpredictable movement patterns of actual elderly subjects over extended periods will provide crucial insights for refining the system's sensitivity and ensuring robustness in clinical settings.



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